MATHEMATICS

GENERAL NOTATION, SETS AND PROBABILITY

General notation

Symbol	Meaning	Examples
$=$	Equal	$2+3=5$ $\quad 4-1=3$
$\neq$	Not equal	$2\neq 3$ $\quad 5+1\neq 7$
$\approx$	Approximately	$\pi\approx 3.14$ $\quad \sqrt{2}\approx 1.41$
$<,\ >$	Less than Greater than	$2<3$ $\quad 5>4$
$\not<,\ \not>$	Not Less than Not Greater than	$2\not<2$ $\quad 4\not>5$
$\leq,\ \geq$	Less/Greater than or equal to	$2\leq 2$ $\quad 5\geq 4$
$\not\leq,\ \not\geq$	Not Less/Greater than or equal to	$3\not\leq 2$ $\quad 4\not\geq 5$
$\pm$	Plus or minus: two choices	$3\pm 1=4$ or 2 $\quad -2\pm 3=1$ or -5
$ab=a(b)=a\cdot b=a\times b$	Multiplication: juxtaposition, parentheses, dot, or cross.	$3x=3\cdot x=3\times x$ $2(4)=2\cdot 4=2\times 4$
$\frac{a}{b}=a/b=a\div b$	Division: fraction, slash, or division sign ($b\neq 0$).	$\frac{12}{3}=12/3=12\div 3=4$ $\frac{x}{5}=x/5\ =x\div 5\ =0.2x$
$\lvert a\rvert$	Absolute value: distance from a to 0	$\lvert -5\rvert=5$ $\quad \lvert 3\rvert=3$ $\quad \lvert 0\rvert=0$
a^2	Squared; multiply a by itself.	$3^2=3\cdot 3=9$ $\quad (-2)^2=(-2)(-2)=4$
a^3	Cubed; multiply three factors of a.	$2^3=2\cdot 2\cdot 2=8$ $\quad (-2)^3=(-2)(-2)(-2)=-8$
$\sqrt{a}=a^{1/2}$	Square root $\quad a\geq 0$	$\sqrt{9}=3$ (not ± 3)
$\sqrt[3]{a}=a^{1/3}$	Cube root $\quad a\in\mathbb{R}$	$\sqrt[3]{-8}=-2$
a^n	Exponentiation: n factors of a	$2^4=2\cdot 2\cdot 2\cdot 2=16$ $\quad (-2)^5=-32$
$\sqrt[n]{a}=a^{1/n}$	nth root.	$\sqrt[4]{16}=16^{\frac{1}{4}}=2$ $\quad \sqrt[5]{-32}=-2=(-32)^{\frac{1}{5}}$

Sets and number systems

Symbol	Meaning	Examples
$\{a,b,c\}$	Set: ignores order and repetition.	$\{2,4\}=\{4,2\}$ $\{2,2,4\}=\{2,4\}$
$\in$	Element of	$2\in\{1,2\}$
$\notin$	Not an element of	$3\notin\{1,2\}$
$\{x:p(x)\}$	$x\in A$ such that $p(x)$ holds.	$\{x\in\mathbb{R}:x>0\}$ Positive real numbers
$\varnothing$	Empty set: no elements.	$\lvert\varnothing\rvert=0$
$\lvert A\rvert$	Number of elements in finite A.	$\lvert\{2,4,6\}\rvert=3$
$A\subseteq B$	Subset: all of A in B; may equal B.	$\{1,2\}\subseteq\{1,2\}$
$A\subsetneq B$	Proper subset: $A\subseteq B$, $A\neq B$.	$\{1\}\subsetneq\{1,2\}$
$A\cup B$	Union: in either or both.	$\{1\}\cup\{2\}=\{1,2\}$
$A\cap B$	Intersection: in both.	$\{1,2\}\cap\{2\}=\{2\}$
$A\setminus B$	Difference: in A, outside B.	$\{1,2\}\setminus\{2\}=\{1\}$
$\mathbb{N}$	Naturals	$\mathbb{N}=\{1,2,3,\ldots\}$
$\mathbb{Z}$	Integers	$Z=\{\ldots,-2,-1,0,1,2,\ldots\}$
$\mathbb{Q}$	Rationals: p/q; $p,q\in\mathbb{Z}$, $q\neq 0$.	$\frac{1}{2},-3\in\mathbb{Q}$
$\mathbb{R}$	Reals: the number line.	$\sqrt{2},\pi,\frac{1}{2},-3\in\mathbb{R}$
$\mathbb{R}\setminus\mathbb{Q}$	Irrationals	$\sqrt{2},\pi\in\mathbb{R}\setminus\mathbb{Q}$
(a,b)	Open: excludes endpoints.	$x\in(1,3):1<x<3$
$[a,b]$	Closed: includes endpoints.	$x\in[1,3]:1\leq x\leq 3$
$[a,b)$	Includes left endpoint only.	$2\in[2,5),5\notin[2,5)$
$(a,b]$	Includes right endpoint only.	$2\notin(2,5],5\in(2,5]$
∞	Unbounded above.	$[0,\infty)$
$-\infty$	Unbounded below.	$(-\infty,0]$

Intervals assume $a<b$. Infinity is not a real number; use parentheses there. Some texts include 0 in $\mathbb{N}$; check the convention. Dots: see Poster 2. Bars: absolute value for numbers, cardinality for finite sets.

Probability and events

Example contexts
Fair die: $\Omega=\{1,2,3,4,5,6\}$.
$A=\{1,3,5\}$ (odd), $B=\{1,2\}$, $C=\{2,4,6\}$.
Die examples below use these events.
Independent fair coin tosses: E: first is heads; F: second is heads.

Symbol	Meaning	Examples
Ω	Sample space: all possible outcomes.	$\Omega=\{1,2,3,4,5,6\}$
ω	Outcome: one possible result.	$\omega=3\in\Omega$
$A\subseteq\Omega$	Event: a set of outcomes.	$A=\{1,3,5\}$
$P(A)$	Probability of A: between 0 and 1.	$0\leq P(A)\leq 1$ $P(\Omega)=1$ $P(\varnothing)=0$
$\frac{\lvert A\rvert}{\lvert\Omega\rvert}$	Equals $P(A)$ for finite, equally likely outcomes only.	$P(A)=\frac{3}{6}=\frac{1}{2}$
A^c	Complement: outcomes in Ω outside A.	$A^c=\{2,4,6\}$ $P(A^c)=1-P(A)=\frac{1}{2}$
$A\cup B$	Event A or B, or both.	$A\cup B=\{1,2,3,5\}$
$A\cap B$	Events A and B together.	$A\cap B=\{1\}$
$A\cap C=\varnothing$	Mutually exclusive: cannot occur together.	$A\cap C=\varnothing$
$P(A\mid B)$	Conditional probability: A given B; $P(B)>0$.	$P(A\mid B)=\frac{P(A\cap B)}{P(B)}$ $=\frac{1/6}{2/6}=\frac{1}{2}$
$P(E\cap F)$	Independent events: multiply their probabilities.	$P(E\cap F)=P(E)P(F)$ $=\frac{1}{2}\cdot\frac{1}{2}=\frac{1}{4}$
$P(A\mid B)$	Independence iff this equals $P(A)$; $P(B)>0$.	$P(E\mid F)=P(E)=\frac{1}{2}$

Keep the ideas separate.
Mutually exclusive events of positive probability are not independent: one occurring rules out the other.

The bar in $P(A\mid B)$ means "given"; in set-builder notation it means "such that". Context determines the reading.

MATHEMATICS

FUNCTIONS, VARIABLES, SEQUENCES AND SUMS

Function notation

Example contexts
Example: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$.
Set and interval notation: see Poster 1.

Symbol	Meaning	Examples
f	Function: one output per domain input.	$f(x) = x^2$
$f(x)$	Value at x; parentheses mark input, not multiplication.	$f(3) = 3^2 = 9$
$f : A \to B$	Maps domain A to codomain B.	$f : \mathbb{R} \to \mathbb{R}$
A	Domain: allowed inputs.	$A = \mathbb{R}$
B	Codomain: declared output target.	$B = \mathbb{R}$
$x \mapsto f(x)$	Maps values; $\to$ connects sets.	$x \mapsto x^2$ $3 \mapsto 9$
$\mathrm{dom}(f)$	Domain: allowed inputs.	$\mathrm{dom}(f) = \mathbb{R}$
$\mathrm{range}(f)$	Range: attained outputs; within codomain.	$\mathrm{range}(f) = [0, \infty)$ No negative outputs.

Letters and Greek symbols

Symbol	Name / description
$a, b, c;\ x, y, z$	Common lowercase Latin letters
A, B, C	Common uppercase Latin letters
α	Alpha
β	Beta
γ	Gamma
δ	Delta
Δ	Uppercase delta
θ	Theta
Θ	Uppercase theta
λ	Lambda
Λ	Uppercase lambda
μ	Mu
σ	Sigma
Σ	Uppercase sigma
ω	Omega
Ω	Uppercase omega (Poster 1)
π	Pi; here, the circle constant $\pi \approx 3.14$.

Names do not fix meanings. Σ is a letter; $\sum$ is the related summation operator (right), not an ordinary variable.

Variables

Letter roles
Roles depend on context: uppercase often names sets; Greek often names angles or numbers. Conventions are not permanent types.

Symbol	Meaning	Examples
x	Variable: may vary in context.	$x \in \{-2, 0, 3.5\}$
m, b	Constants: fixed in context.	$y = mx + b$
x	Input for a line.	$y = mx + b$
y	Output for that line.	m, b fixed
n	May count objects.	$n \in \mathbb{N}$
θ	May name an angle in degrees ($^\circ$).	$\theta = 30^\circ$
A	May name a set.	$A = \{1, 2\}$
λ	May name a number.	$\lambda = 2$

A letter may be variable or constant. Values -2, 0, 3.5 are separate choices for x, not simultaneous equalities.

Lists, sequences and subscripts

Reading guide
A **sequence** is an ordered list of terms; a **series** is a sum of sequence terms. Here, parentheses show sequences; braces show sets.
Sequence examples: $a_i = 2i$, starting at $i = 1$.

Symbol	Meaning	Examples
$\ldots$	Omitted list entries.	$1, 2, 3, \ldots$
$\cdots$	Omitted terms in a sum.	$a_1 + a_2 + \cdots + a_n$
$1, 2, \ldots, n$	Finite list: ends at n.	$1, 2, 3, \ldots, 5$ $(n = 5)$
$1, 2, 3, \ldots$	Infinite list: no final term.	$1, 2, 3, 4, 5, \ldots$
a_i	Term at position or label i.	$a_3 = 2(3) = 6$
a_1	Term at index 1.	$a_1 = 2$
a_n	Term at index n.	$a_n = 2n$
$(a_i)_{i=1}^{4}$	Finite: four terms.	$(a_i)_{i=1}^{4} = (2, 4, 6, 8)$
$(a_i)_{i=1}^{\infty}$	Infinite: no last term.	$(a_i)_{i=1}^{\infty} = (2, 4, 6, \ldots)$
$(2, 4)$	Order matters.	$(2, 4) \neq (4, 2)$
$(2, 2, 4)$	Repetition matters.	$(2, 2, 4) \neq (2, 4)$
A_1, A_2	Subscripts may label cases.	Alternatives A_1 and A_2.

Dots follow an established pattern; alone they do not define a unique rule. The finite list $1, 2, 3, \ldots, n$ assumes integer $n \geq 3$. An index is not multiplication or an exponent. Indices may start at 1, 0, or another integer: state the start. Sets ignore order and repetition (Poster 1).

Summation notation

Index convention
Indices run through consecutive integers between the bounds, **inclusive**. Here $n \in \mathbb{N}$, $n \geq 1$, and $a_i = 2i$ in the sequence examples.

Symbol	Meaning	Examples
$\sum_{i=1}^{n} a_i$	Sum of the indicated terms.	$a_1 + a_2 + \cdots + a_n$
i	Index: local to sum.	$i = 1, 2, \ldots, n$
a_i	Summand: expression summed.	$a_i = 2i$
1	Lower bound: first index.	Start at $i = 1$.
n	Upper bound: last index.	End at $i = n$.
a_n	Single term at n.	$a_4 = 8$
S_n	Sum of first n terms; start at 1.	$S_n = \sum_{i=1}^{n} a_i$
S_4	Four terms for $a_i = 2i$.	$S_4 = 2 + 4 + 6 + 8 = 20$
$\sum_{i=2}^{4} 3$	Constant; bounds 2 to 4.	$\sum_{i=2}^{4} 3 = 3 + 3 + 3 = 9$
$\sum_{k=1}^{3} k^2$	Squares; bounds 1 to 3.	$\sum_{k=1}^{3} k^2 = 1 + 4 + 9 = 14$
$i,\ k$	Index names may change consistently.	$\sum_{i=1}^{4} i = \sum_{k=1}^{4} k$ $= 1 + 2 + 3 + 4 = 10$

Rename the local index in both bounds and summand. Centered dots use the same omission convention as list dots.